Pipeline Workflow Diagrams

1. Simple Legacy Workflow

```
flowchart LR
    A[Source System / Input Files] --> B[Matillion Extract Job]
    B --> C[RAW Landing Tables]
    C --> D[Legacy Transformation Job]
    D --> E[DL_LEGACY / Legacy Output Layer]
    E --> F[Validation Checks]
    F --> G[Demo Consumption / Reporting]

    H[Run Control] --> B
    H --> D
    I[Error Log / Audit] --> D
    D --> I
```

Workflow Notes

- Raw data is loaded first into landing tables.
- Transformation logic is handled in the legacy pipeline layer.
- Validation is executed after transformation before demo consumption.
- This flow is used as the baseline for future-state comparison.

2. Simple Future Workflow

```
flowchart LR
    A[Source System / Input Files] --> B[Ingestion / EL Job]
    B --> C[RAW Snowflake Layer]
    C --> D[dbt Source Models]
    D --> E[STAGING Models]
    E --> F[CURATED / Intermediate Models]
    F --> G[DW / Final Mart]
    G --> H[Tests + Documentation + Lineage]
    H --> I[Demo Consumption / Reporting]

    J[Scheduler / Orchestrator] --> B
    J --> D
    K[Freshness + Quality Checks] --> E
    K --> F
    K --> G
```

Workflow Notes

- EL loads data into RAW, while dbt owns transformations.
- The transformation path is modular and easier to explain.
- Tests and lineage are part of the workflow, not an afterthought.
- This becomes the target-state reference model.

3. Moderate Legacy Workflow

```
flowchart TB
    A[Source 1: Operational Data] --> D[Matillion Extract & Load]
    B[Source 2: Planning / Reference Data] --> D
    C[Source 3: Flat Files / Mapping Tables] --> D

    D --> E[RAW Landing Zone]
    E --> F[Legacy Join / Mapping Logic]
    F --> G[Business Rules / Calculations]
    G --> H[Aggregated Legacy Outputs]
    H --> I[DL_LEGACY / Reporting Tables]
    I --> J[Validation + Reconciliation]
    J --> K[Demo Reports / Walkthrough]

    L[Run Control] --> D
    L --> F
    L --> G
    M[Audit / Error Logging] --> F
    G --> M
```

Workflow Notes

- Multiple data sources are integrated before transformation.
- Business logic and mapping are embedded in the legacy workflow.
- Aggregation and final reporting outputs are prepared in the legacy layer.
- This version helps demonstrate complexity in the old model.

4. Moderate Future Workflow

```
flowchart TB
    A[Source 1: Operational Data] --> D[EL Ingestion]
    B[Source 2: Planning Data] --> D
    C[Source 3: Flat Files / Reference Data] --> D

    D --> E[RAW Snowflake Layer]
    E --> F[dbt Sources]
    F --> G[STAGING Models\ncleaning / typing / standardization]
    G --> H[Intermediate Models\njoins / mappings / rule sets]
    H --> I[CURATED Business Models]
```

```
    I --> J[DW Facts & Dimensions]
    J --> K[Tests / Docs / Lineage / Reconciliation]
    K --> L[Demo Reports / Comparison Layer]

    M[Orchestration Layer] --> D
    M --> F
    N[Source Freshness Checks] --> F
    O[Data Quality Tests] --> G
    O --> H
    O --> I
    O --> J
```

Workflow Notes

- Multiple sources are ingested, standardized, and transformed in layers.
- dbt separates staging, integration, business logic, and final marts.
- Quality checks, lineage, and reconciliation are built into the process.
- This flow is ideal for explaining migration and comparison against the legacy state.

5. Comparison Structure

```
flowchart LR
    A[Simple Legacy Output] --> E[Comparison Layer]
    B[Simple Future Output] --> E
    C[Moderate Legacy Output] --> F[Comparison Layer]
    D[Moderate Future Output] --> F
    E --> G[Row Count / Metric Validation]
    F --> H[Business Rule Validation]
    G --> I[Demo Narrative]
    H --> I
```

Comparison Notes

- Keep outputs aligned by business entity and metric.
- Use the same validation pattern for all four workflows.
- This structure makes later comparison simple and repeatable.